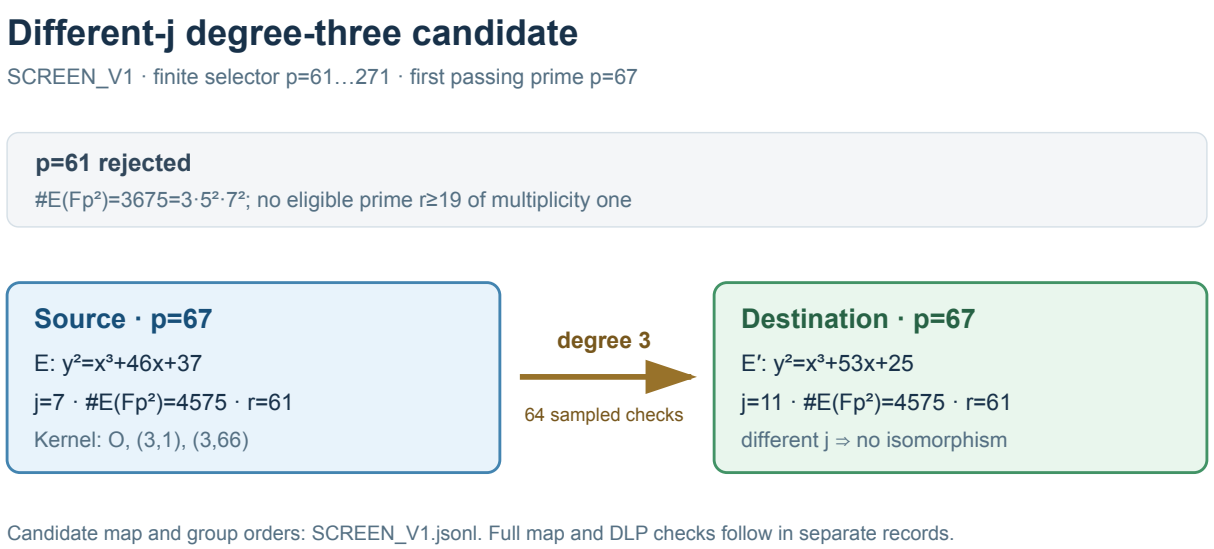
Different-j degree-three candidate screen

Run: SCREEN_V1, 2026-10-15. **Status:** first selector pass at $p = 67$. **Host:** shared Darwin, L0.

The frozen selector started with $E : y^2 = x^3 - 21x + 37$ and rational $T = (3, 1)$, searched ascending primes in $[61, 271]$ with $p = 1 \bmod 3$, and required a multiplicity-one subgroup prime $r \geq 19$. At $p = 61$, $n_2 = 3675 = 3 \cdot 5^2 \cdot 7^2$ had no eligible factor. At $p = 67$, both curves had $n_2 = 4575 = 3 \cdot 5^2 \cdot 61$, $r = 61$, and cofactor 75. Source coefficients were $(46, 37)$ with $j = 7$; destination coefficients were $(53, 25)$ with $j = 11$.



The implemented Vélu sum map had kernel $\{O, (3, 1), (3, 66)\}$ and passed 64 sampled on-curve and homomorphism checks. Different j-values exclude an isomorphism for the nonsingular pair; the map's full-point check and DLP transfer are separate pending gates. The raw record is `../../SCREEN_V1.jsonl`, SHA-256 `be2094594f2ca0e7ec49d221a8d9865b5d5ecf640922cd0541520f414ac29707`. The command, source hash, and source link are in `REPORT.md` beside this PDF. See the MIT 18.783 lecture slides on j-invariants for the isomorphism invariant.